

Pritam Banerjee

Post-doctoral researcher, University of California, Berkeley, California, USA

PROFESSIONAL EXPERIENCE

University of California, Berkeley, USA

09/01/2023-Present

Postdoctoral Researcher (PIs: Profs. Rosemary Gillespie and George K. Roderick)

Project: Next Generation Biosecurity Monitoring of Invasive Alien Arthropod Species with eDNA

Basirhat College, West Bengal, India

01/01/2019-08/01/2019

Guest Lecturer

Undergraduate teaching: Ecology, Genetics, DNA metabarcoding, Molecular Biology

Chandraketugarh Sahidullah Smriti Mahavidyalaya, West Bengal, India

06/01/2018-01/01/2019

Guest Lecturer

Undergraduate teaching: Ecology, Genetics, DNA metabarcoding, Molecular Biology

EDUCATION

National Chung Cheng University, Chiayi County, Taiwan

10/02/2019-08/10/2023

PhD in Molecular Biology (Industry-Academia Project) (PI: Prof. Chien- Yen Chen)

(Environmental DNA; DNA metabarcoding, Invasion ecology, Population genetics, Bioinformatics)

National Chung Cheng University, Chiayi County, Taiwan

10/02/2019- 01/01/2021

Diploma in Science, Technology, Engineering, and Mathematics (STEM)

(Environmental Science, Biotechnology, Material Science, Molecular Biology)

Kalyani University, West Bengal, India

06/02/2016- 01/01/2021

MSc in Botany

(DNA barcoding; Genetics, Plant taxonomy, Plant ecology, Plant physiology, Ecology and evolution)

West Bengal State University, West Bengal, India

06/20/2013- 05/20/2026

BSc in Botany

(Major: Botany; Minor: Zoology, Chemistry)

(DNA barcoding; Genetics, Plant taxonomy, Plant ecology, Plant physiology, Ecology and evolution)

Basirhat High School, West Bengal, India

2001-2013

High School

(Sub: Biology, Physics, Chemistry, Mathematics, Language: Bengali, English)

TEACHING AND MENTORSHIP

Lecturer

- **Basirhat College, West Bengal State University (WBSU)**, West Bengal, India. [Courses: Integrated Life Sciences (BOTPCOR01T), Plant Ecology and Environmental Biology (BOTPCOR07T), and Molecular & Cellular Genetics (BOTPCOR11T)]
- **Chandraketugarh Sahidullah Smriti Mahavidyalaya, West Bengal State University (WBSU)**, West Bengal, India. (Courses: Plant Ecology and Taxonomy- paper II and Biodiversity and Conservation-ZOODSC717T)
- **Hooghly Mohsin College, University of Burdwan (BU)**, West Bengal, India (Post Graduate Department of Botany). (Course: Biostatistics)

Supervisor and Mentorship

National Chung Cheng University, Taiwan: Nalonda Chatterjee and Chin-Wen Wang (Master's, National Chung Cheng University, Taiwan); TEEP Interns: Chanon Buntham, Chayada Jaidee, Thanyalak Kammeerak (National Chung Cheng Univ., Taiwan).

UC Berkeley URAP students: Mackenzie Lasher (Sept 2023–Present); Mia Di Genova, Jonathan Gasser-Brenna, Melissa Andico Murphy (Sept–Dec 2023); Rosemary Gomes, Lina Wolksi, Sophia Razavi, Nancy Gutierrez, Ashanti Gamez, Christopher Kim, Azka Khan (Jan–May 2024); Patrick Chang, Lauren Mandel (Jan 2024–Present); Jenna Tran (June 2024–Present).

SELECTED PEER-REVIEWED PUBLICATIONS

(* = corresponding author, # = shared first author)

DNA Barcoding, DNA Metabarcoding, Environmental DNA

- **Banerjee, P***, Al - Bayer, S., Calaor, J., Weber, S., Graham, N.R., Andersen, J.C., Economo, E.P., Kennedy, S., Krehenwinkel, H., Gillespie, R.G. and Roderick, G.K., 2026. Comparison of environmental DNA and bulk DNA metabarcoding for assessing terrestrial arthropod diversity across three habitat types on Guam. **Molecular Ecology Resources**, 26(5), p.e70172.
- **Banerjee, P***, Chatterjee, N., Chen, J.S., Dey, G., Sharma, R.K., Maity, J.P., Wang, C.W., Jaidee, C., Kammeerak, T., Tangwattanachuleeporn, M. and Poolpol, K., 2025. Consumption of invasive apple snails (*Pomacea* spp.): a public health concern for neuroangiostrongyliasis in Taiwan. **Journal of Environmental, Pharmaceutical and Sustainability Science**, 2(3), pp.1-17.
- **Banerjee, P***, Dey, G., Stewart, K.A., Barnes, M.A., Taharia, M., Seymour, M., Wang, C.W., Sharma, R.K., Maity, J.P. and Chen, C.Y., 2026. Environmental DNA Metabarcoding Effectively Detects Invasive Species, Pests, and Community Changes in Taiwan's Rice Fields. *Ecology and Evolution*, 16(4), p.e73339.
- **Banerjee, P***, Maity, J.P., Chatterjee, N., Weber, S., Dey, G., Sharma, R.K. and Chen, C.Y., 2025. Harnessing Environmental DNA (eDNA) to Explore Frugivorous Interactions: A Case Study in Papaya (*Carica papaya*) and Pineapple (*Ananas comosus*), **Environmental DNA**. (DOI: <https://doi.org/10.1002/edn3.70196>)
- **Banerjee, P.**, Chatterjee, N., Stewart, K.A., Chen, J.S., Dey, G., Maity, J.P., Sharma, R.K., Wang, C.W., Lin, P.Y., Lee, K.H. and Chen, C.Y., 2024. Taiwan's mysterious mollusks: a deep dive into the cryptic hybridization of *Pomacea canaliculata* and *Pomacea maculata*. **Evolutionary Ecology**, 38(6), pp.871-883.
- **Banerjee, P.**, Dey, G., Maity, J.P., Stewart, K.A., Sharma, R.K., Chan, M.W., Lee, K. and Chen, C.Y., 2024. The unseen invaders: Tracking phylogeographic dynamics and genetic diversity of cryptic *Pomacea canaliculata* and *P. maculata* (Golden apple snails) across Taiwan. **Ecology and Evolution**, 14(4), p.e11268.
- **Banerjee, P.**, Stewart, K.A., Dey, G., Sharma, R.K., Maity, J.P., Chan, M.W., Chang, K.P., Chen, T.H., Hsu, C.T. and Chen, C.Y., 2022. When conventional methods fall short: identification of invasive cryptic Golden Apple Snails (*Pomacea canaliculata*; *P. maculata*) using environmental DNA. **Hydrobiologia**, 849(19), pp.4241-4257.

- **Banerjee, P.**, Stewart, K.A., Antognazza, C.M., Bunholi, I.V., Deiner, K., Barnes, M.A., Saha, S., Verdier, H., Doi, H., Maity, J.P. and Chan, M.W., 2022. Plant–animal interactions in the era of environmental DNA (eDNA)—a review. *Environmental DNA*, 4(5), pp.987-999.
- **Banerjee, P.**, Stewart, K.A., Dey, G., Antognazza, C.M., Sharma, R.K., Maity, J.P., Saha, S., Doi, H., de Vere, N., Chan, M.W. and Lin, P.Y., 2022. Environmental DNA analysis as an emerging non-destructive method for plant biodiversity monitoring: a review. *AoB Plants*, 14(4), p.plac031.
- **Banerjee, P.**, Dey, G., Antognazza, C.M., Sharma, R.K., Maity, J.P., Chan, M.W., Huang, Y.H., Lin, P.Y., Chao, H.C., Lu, C.M. and Chen, C.Y., 2021. Reinforcement of environmental DNA based methods (sensu stricto) in biodiversity monitoring and conservation: a review. *Biology*, 10(12), p.1223.
- Rodriguez-Ezpeleta, N., Morissette, O., Bean, C.W., Manu, S., **Banerjee, P.**, Lacoursière-Roussel, A., Beng, K.C., Alter, S.E., Roger, F., Holman, L.E. and Stewart, K.A., 2021. Trade-offs between reducing complex terminology and producing accurate interpretations from environmental DNA: Comment on “Environmental DNA: What's behind the term?” by Pawlowski et al.,(2020). *Molecular Ecology*, 30(19), pp.4601-4605.

Ecology, Environment, and Pollution

- Dey, G., Maity, J.P., **Banerjee, P.**, Sharma, R.K., Das, K., Gnanachandrasamy, G., Wang, C.W., Lin, P.Y., Wang, S.L. and Chen, C.Y., 2024. Evaluation and mitigation of potentially toxic elements contamination in mangrove ecosystem: Insights into phytoremediation and microbial perspective. *Marine Pollution Bulletin*, 209, p.117035.
- Mahapatra, S., Maity, J.P., Singha, S., Mishra, T., Dey, G., Samal, A.C., **Banerjee, P.**, Biswas, C., Chattopadhyay, S., Patra, R.R. and Patnaik, S., 2024. Microplastics and nanoplastics in environment: Sampling, characterization and analytical methods. *Groundwater for Sustainable Development*, 26, p.101267.
- Dey, G., Maity, J.P., **Banerjee, P.**, Sharma, R.K., Etesami, H., Bastia, T.K., Rath, P., Sukul, U., Huang, H.B., Huang, K.W. and Chen, C.Y., 2024. Characterization of halotolerant phosphate-solubilizing rhizospheric bacteria from mangrove (*Avicennia* sp.) with biotechnological potential in agriculture and pollution mitigation. *Biocatalysis and Agricultural Biotechnology*, 55, p.102960.
- Dey, G., Patil, M.P., Banerjee, A., Sharma, R.K., **Banerjee, P.**, Maity, J.P., Singha, S., Taharia, M., Shaw, A.K., Huang, H.B. and Kim, G.D., 2023. The role of bacterial exopolysaccharides (EPS) in the synthesis of antimicrobial silver nanomaterials: A state-of-the-art review. *Journal of Microbiological Methods*, 212, p.106809.
- Dey, G., **Banerjee, P.**, Maity, J.P., Sharma, R.K., Gnanachandrasamy, G., Huang, Y.H., Huang, H.B. and Chen, C.Y., 2022. Heavy metals distribution and ecological risk assessment including arsenic resistant PGPR in tidal mangrove ecosystem. *Marine Pollution Bulletin*, 181, p.113905.
- Dey, G., **Banerjee, P.**, Sharma, R.K., Maity, J.P., Etesami, H., Shaw, A.K., Huang, Y.H., Huang, H.B. and Chen, C.Y., 2021. Management of phosphorus in salinity-stressed agriculture for sustainable crop production by salt-tolerant phosphate-solubilizing bacteria—A review. *Agronomy*, 11(8), p.1552.

PEER-REVIEWED BOOK CHAPTER

- Maity, J.P., Huang, Y.H., Day, G., **Banerjee, P.**, Samal, A.C., Ahmad, A., Bhattacharya, P. and Chen, C.Y., 2023. Arsenic transformation and possible mobilization by indigenous microbes in hot spring environment. In *Arsenic in the Environment: Bridging Science to Practice for Sustainable Development As2021* (pp. 61-62). CRC Press.

SUBMITTED PAPERS

- Weber, S., Hutchins, L., **Banerjee, P.**, Callaghan, W., Farrow, A.A., Andersen, J., Gillespie, R. and Roderick, G.K., 2026. Topography structures of arthropod communities revealed by leaf-derived environmental DNA on O‘ahu, Hawai‘i. *bioRxiv*, pp.2026-04 (Under review in Environmental DNA).
- **Weber, S.**, **Banerjee, P.**, Scali, E., Farrow, A.A., Boren, A.M., Russelk III, W.T., Gillespie, R., Graham, N.R. and Roderick, G., 2026. Extraction workflow determines marker-specific recovery and reproducibility in leaf-litter eDNA metabarcoding. *bioRxiv*, pp.2026 (Under review in Methods in Ecology & Evolution).

CURRENT PROJECTS

- **Maximizing arthropod biodiversity monitoring:** A Comparative analysis of environmental DNA metabarcoding with DNA metabarcoding across different forest types in Guam.
- **Invasive Species Detection:** A Hybrid Approach Using Traditional and eDNA Tools/methods for/in Island Ecosystems.
- **Understanding the shape of species interactions with environmental DNA** among native and introduced species in Hawaii with swabbing eDNA
- **Comparative analysis of traditional sampling and eDNA-based methods** to detect Hawaiian arthropods from natural and disturbed gradients.
- **Spider web as a natural eDNA sampler:** A case study to detect air eDNA from the Hawaiian Islands
- **Reference database of Pacific Islands:** Hawaii and Guam

CONFERENCES

Organizing Committee

- “Arsenic and Other Pollutants, Water Security and One Health under Global Climate Change Scenario”, Kalinga Institute of Industrial Technology- Deemed to be University (KIIT-DU), Odisha, India.

Oral Presentation

- Banerjee, P., et al., 2026, “*Plant-Animal Interactions in the era of environmental DNA*” Daru lab, **Stanford University, Stanford, USA.**
- Banerjee, P., et al., 2025, “Spider webs as a natural eDNA sampler across the Hawaiian islands” **Ecolunch, University of California, Berkeley, USA.**
- Banerjee, P., et al., 2024, “DNA-based solution to the biodiversity monitoring.” **Essig Museum talk, University of California, Berkeley, USA.**
- Banerjee, P., et al., 2024, Can we detect plant-animal interactions with environmental DNA? **Botany lunch, University of California, Berkeley, USA.**
- Banerjee, P., et al., 2024. Comparison of environmental DNA and bulk DNA metabarcoding for assessing terrestrial arthropod diversity across three habitat types on Guam, **The International Barcode of Life project (iBOL), Belém, Brazil.**
- Banerjee P. *, Dey G., Maity, J.P., Chatterjee N., Sharma R.K., Taharia M., Wang C.W., Chen, C.Y. *Assessing Heavy Metal Contamination and Biodiversity through eDNA in Conventional vs. Organic Rice Fields for Sustainable Agriculture. 9th International Congress & Exhibition on Arsenic in the Environment 2024*, Odisha, India.
- Banerjee P. *, Dey G., Sharma R.K., Maity, J.P., Chen, C.Y., 2023. *Revealing the potential for environmental DNA (eDNA) to detect species interactions in two economically important plants: Carica papaya and Ananas comosus.* The eDNA Society International Meeting 2023, Otsu City, Shiga, **Japan.**
- Banerjee P. *, Dey G., Sharma R.K., Maity, J.P., Chen, C.Y., 2023. *Urgent action needed: combatting the menace of invasive Pomacea. III International Young Researchers Conference on Invasive Species*; Pages 60 – 61, ISBN: 978-84-09-51676-6.
- Banerjee P. *, Dey G., Sharma R.K., Chen, C.Y., Huang Y.H., Lin P.Y., 2021. *Environmental DNA over conventional taxonomy: eliminating false biodiversity information in case of cryptic invasion.* The 26th Symposium of Young Asian Biological Engineers’ Community.
- Banerjee, P. *, Stewart, K.A., Dey, G., Sharma, R.K., Maity, J.P., Chan, M.W., Chang, K.P., Chen, T.H., Hsu, C.T. and Chen, C.Y., 2022. *When conventional methods fall short: identification of invasive cryptic Golden Apple Snails (Pomacea canaliculata; P. maculata) using environmental DNA.* Emerging Technologies Information Sessions

(ETIS) 2022, Pacific Northwest Aquatic Monitoring Partnership a forum for coordination and collaboration, <https://www.pnamp.org/project/emerging-technologies-information-sessions-2022>.

- **Banerjee, P. ***, Dey, G., Sharma, R.K., Maity, J.P., and Chen, C.Y., 2021. *Applications of novel Environmental DNA(eDNA) technique in detection of invasive species and environmental monitoring*, 3rd Ecology and Evolution Ireland Conference 7-8th January 2021.
- **Banerjee P. ***, Dey G., Sharma R.K., Anootha P.N.V., Maity, J.P., Chen, C.Y., 2020. *Environmental DNA based detection and monitoring of highly invasive Green iguana (Iguana iguana) in Taiwan*. The International Conference on Environmental and Civil Engineering Innovation 2020 (ICECEI2020), Taichung, Taiwan.

Invited talks

- **Banerjee et al.**, “Next Generation Biosecurity Monitoring of Invasive Alien Arthropod Species on Pacific Islands”, Zhejiang Provincial Key Laboratory of Biometrology and Inspection & Quarantine, College of Life Sciences, **China Jiliang University, Hangzhou, China**.
- **Banerjee et al.**, “Molecular Approaches to Understanding the Invasion Dynamics of Cryptic Golden Apple Snails (Pomacea spp.) in Taiwan: A Focus on Environmental DNA and DNA-Based Techniques”, Zhejiang Provincial Key Laboratory of Biometrology and Inspection & Quarantine, College of Life Sciences, **China Jiliang University, Hangzhou, China**.
- **Banerjee et al.**, “Environmental DNA based biodiversity monitoring: Advantages, disadvantages and future recommendation” YEN group, Department of Earth and Environmental Sciences, **National Chung Cheng University, Chiayi County, Taiwan**.

VISITING SCHOLAR

- **China Jiliang University**, Zhejiang, Hangzhou, China. PI: **Qianqian Yang**
- **National Chung Cheng University**, Chiayi County, Taiwan. PI: **Chien- Yen Chen**
- **Kalinga Institute of Industrial Technology (KIIT) University**, Odisha, India. PI: **Jyoti Prakash Maity**

INTERNSHIP

- **Genomics BioScience Tech. Co., Ltd, Taiwan** “Internship program on “Bioinformatics and laboratory-based technique for Next Generation Sequencing” (July 2022-Aug 2022).

GRANTS AND FELLOWSHIPS

- Postdoctoral funding from Strategic Environmental Research and Development Program (SERDP), USA
- Awarded 4 years of grant for Ministry of Education (MOE)- Industry-Academic Project (Taiwan), No 110CF59-02 (June 2019-June 2024): Total amount 0.8 million NTD.
- Awarded 2-year Overseas Research Scholarships (ORS; A503) from National Chung Cheng University, Taiwan (March 2021-March 2023): Total amount 0.4 million NTD.
- Awarded the Swami Vivekananda merit cum means scholarship in India.

PROFESSIONAL SERVICES

Peer-reviewer

Nature Communications; Biological Conservation; Environmental DNA; Acta Ecologica Sinica; Biological Invasions; Scientific Reports; Management of Biological Invasions; Current Opinion in Environmental Science & Health; Biodiversity and Conservation; Applications in Plant Sciences; Ecological Indicators; etc.

SKILLS

Technical Skills

- **DNA Extraction & Purification:** Extensive experience with diverse environmental samples (soil, water, air, plant surfaces, leaf litter, spider webs, etc.), as well as plant and animal DNA, using commercial kits and self-developed cost-efficient protocols.
- **Environmental Parameter Analysis:** Skilled in assessing water, soil, and air quality.
- **Field Survey & Sampling:** Experienced in long-term field surveys, including driving, camping, and sample collection from remote and inaccessible areas.
- **Polymerase Chain Reaction (PCR):** Proficient in conventional PCR, qPCR, and droplet digital PCR (ddPCR); expertise in primer design, PCR product purification, index PCR, pooling, and fragment analysis.
- **Automated Laboratory Systems:** Experienced in handling semi-automated and fully automated DNA extraction and PCR preparation machines.
- **DNA Barcoding & Metabarcoding:** Hands-on experience with Sanger sequencing, Illumina sequencing, and Nanopore sequencing (from sample preparation to sequencing).
- **Molecular Cloning & RFLP:** Skilled in restriction fragment length polymorphism (RFLP), cloning, and plasmid preparation.
- **Microscopy:** Proficient in light microscopy, fluorescence microscopy, scanning electron microscopy (SEM), and transmission electron microscopy (TEM).
- **Electrophoresis Techniques:** Expertise in gel electrophoresis and capillary electrophoresis; well-trained for large-scale projects with strong troubleshooting abilities.
- **Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES):** Experienced in heavy metal analysis.
- **Microbial Culture & Identification:** Skilled in culturing and identifying bacteria and fungi.

Computational & Data Analysis Skills

- **DNA sequence analysis:** BioEdit, MEGA X, CLUSTAL W, Geneious.
- **Phylogenetics & Evolutionary Analysis:** BEAST2, MrBayes, FastTree, jModelTest, FigTree, iTOL, Tracer
- **Population Genetics & Genetic Diversity Analysis:** PopART, DnaSP v6, STRUCTURE
- **DNA Metabarcoding & Microbiome Analysis:** QIIME2, USEARCH, VSEARCH, Cutadapt, FastQC, Fastp, DADA2, Galaxy
- **Geospatial & Environmental Analysis:** ArcGIS, QGIS
- **General Bioinformatics & Data Analysis:** Linux, R, Origin 10, Microsoft Excel
- **AI tools:** ChatGPT, Litmaps, ResearchRabbit, Connected Paper
- **Image Analysis & Visualization:** ImageJ, Inkscape, PowerPoint, Photoshop, Biorender, ChemDraw
- **Scientific Writing & Research Communication:** MS Word, Google Docs, Collanote, Mendeley, Paper Digest, TheQuillBot.
- **Bioinformatics Databases & Resources:** NCBI, Nephele, Bold

LANGUAGE SKILLS

English; Bengali; Hindi; Chinese (Basic)

HOBBIES

Hiking and Camping; Wildlife photography; Physical fitness; Cricket; Sci-Fi story reading

REFERENCES

Prof. Rosemary Gillespie, Professor at the Department of Environmental Science, Policy, & Management University of California, Berkeley, USA. Email: gillespie@berkeley.edu; Ph: +1 (510) 717-3445. (Postdoctoral Advisor)

Prof. George K. Roderick, Professor at the Department of Environmental Science, Policy, & Management University of California, Berkeley, USA. Email: roderick@berkeley.edu; Ph: +1 (510) 717-3295. (Postdoctoral Advisor)

Prof. Chien- Yen Chen, Prof and Dean of the College of Science, National Chung Cheng University, Taiwan. Email : chenchienyen@gmail.com/chien-yen.chen@oriel.oxon.org, Ph: +886937765016. (PhD Advisor)

Prof. Jyoti Prakash Maity, Professor, Department of Chemistry, School of Applied Sciences, KIIT Deemed to be University, Bhubaneswar, Odisha. Email: jyoti_maity@yahoo.com, Ph: +917363988133. (PhD Advisor)

Prof. Zahed Hossain, Department of Botany, University of Kalyani, West Bengal, India. Email: zahed_klv@yahoo.com, Ph: +918697901534. (MSc Advisor).